BILINEAR VARIATION FROM THE HEAT EQUATION

Abstract. As a consequence of our variation estimates and Calderón's transference principle, we resolve the longstanding open problem of pointwise convergence of bilinear ergodic averages associated with two commuting measure-preserving transformations in the full Banach range $L^{p_1} \times L^{p_2} \to L^p$, where $1 \le p_1, p_2 \le \infty$ and $1/p = 1/p_1 + 1/p_2 \le 1$. We prove the Gaussian r-variation estimate from $L^3 \times L^3$ to $L^{3/2}$ for $r > 2$, and the lacunary sharp-average estimate for $r > 3$. The heat chain rule and one Cauchy–Schwarz inequality bound the required scale integral by the decrease of a positive trace energy. A scalar identity for Gaussian differences produces one continuously differentiable coefficient path and the same Gaussian derivative in all three variables. The proof works with general L^3 inputs throughout.

1. Pointwise convergence for two commuting transformations

The estimates below yield the following resolution of the longstanding pointwise convergence problem for two commuting transformations.

Theorem 1.1. *Let $(X, \mathcal{B}, \mu)$ be a probability space, and let S, T be commuting invertible measure-preserving transformations of X. Suppose that $1 \le p_1, p_2 \le \infty$ and*

$$\frac{1}{p} = \frac{1}{p_1} + \frac{1}{p_2} \le 1, \qquad \frac{1}{\infty} = 0.$$

For every $f \in L^{p_1}(X)$ and $g \in L^{p_2}(X)$, the averages

$$A_N^{S,T}(f,g)(x) = \frac{1}{N} \sum_{n=0}^{N-1} f(S^n x) g(T^n x)$$

converge for μ-almost every x as $N \to \infty$. Their limit belongs to $L^p(X)$.

The connection with the Euclidean estimates proved below is provided by Calderón's transference principle.

2. The theorem and the estimate behind it

For $q \in L^1(\mathbb{R})$ write $q_t(u) = t^{-1} q(u/t)$ and

$$T_t^q(F,G)(x,y) = \int F(x+u, y) G(x, y+u) q_t(u)\, du.$$

Put $\gamma(u) = e^{-\pi u^2}$, $\mathcal{G}_t = T_t^{\gamma}$ and $\mathcal{A}_t = T_t^{\mathbf{1}_{[0,1]}}$. For an ordered sequence define the homogeneous variation by

$$V^r(z_n : n) = \sup_{n_0 < \cdots < n_m} \left(\sum_{j=1}^{m} |z_{n_j} - z_{n_{j-1}}|^r \right)^{1/r}.$$

Throughout, $t_{n+1} \ge \lambda t_n$ for a fixed $\lambda > 1$.

Theorem 2.1. *If $r > 2$ and $F, G \in L^3(\mathbb{R}^2)$, then*

$$\|V^r(\mathcal{G}_{t_n}(F,G) : n)\|_{3/2} \le C \frac{r-1}{r-2} \|F\|_3 \|G\|_3.$$

The absolute constant is uniform over finite lists and independent of the lacunarity ratio. The Gaussian proof in fact uses only the ordering of the times.

Theorem 2.2. *For $r > 3$, $\|V^r(\mathcal{A}_{t_n}(F,G) : n)\|_{3/2} \le C_{\lambda,r}\|F\|_3\|G\|_3$.*

The sharp comparison is proved separately in Appendix B. Its interpolation requires $r > 3$. The main proof uses cancellation in the curvature expressions, whose scale integrals are bounded by telescoping, following the strategy of Durcik–Kovač–Škreb–Thiele [1].

Let ρ_s denote the centred Gaussian density of variance s. Fix $0 < t_0 < \cdots < t_N$. For complex coefficients $c_1, \dots, c_N$ put

$$c_0 = c_{N+1} = 0, \qquad H = \max_n |c_n|, \qquad V = \sum_{n=0}^{N} |c_{n+1} - c_n|, \tag{2.1}$$

and, with $\langle X, c\rangle = \int X\bar{c}$, define

$$\Lambda_\gamma(F,G;c) = \sum_{n=1}^{N} \langle (\mathcal{G}_{t_n} - \mathcal{G}_{t_{n-1}})(F,G), c_n\rangle.$$

The endpoint changes ensure $V \ge 2H$.

Theorem 2.3 (Coefficient estimate)**.** *For $F, G, H, V \in L^3(\mathbb{R}^2)$,*

$$\boxed{|\Lambda_\gamma(F,G;c)| \le C\|F\|_3\|G\|_3\sqrt{\|H\|_3\|V\|_3}.} \tag{2.2}$$

The constant is independent of the number and separation of the times. The bound uses $\|H\|_3$ and $\|V\|_3$ separately.

This is the only analytic estimate needed for Gaussian variation. We first give its variational consequence, prove the Gaussian energy inequality, and then relate it to the original pairing and estimate the coefficient derivative.

3. From coefficients to variation

Fix a finite list, put $b_n = \mathcal{G}_{t_n}(F,G)$, and let $q = r' < 2$. Hölder at each translation and $\gamma \in L^1$ give $b_n \in L^{3/2}$, so its finite-list variation also lies in $L^{3/2}$ before any uniform bound is proved. Let $\mathcal{P}$ be its finite collection of increasing chains $\pi = (n_0 < \cdots < n_m)$, and write $z_{\pi,j} = b_{n_j} - b_{n_{j-1}}$. Ordinary finite-dimensional vector-valued duality gives

$$\|V^r(b_n : n)\|_{3/2} = \sup_{\|W\|_3 \le 1} \left| \sum_{\pi,j} \langle z_{\pi,j}, w_{\pi,j}\rangle \right|, \qquad W = \sum_\pi W_\pi, \quad W_\pi = \left(\sum_j |w_{\pi,j}|^q \right)^{1/q}. \tag{3.1}$$

Indeed the finite sequence space is the ℓ^∞ sum over chains of their ℓ^r spaces; its dual is the corresponding ℓ^1 sum of ℓ^q spaces. Combining this norm identity with $L^{3/2}$–L^3 duality proves (3.1). Thus no spatially selected chain needs to be introduced.

For $0 < u < 1$, put $\omega_{\pi,j} = w_{\pi,j}/|w_{\pi,j}|$ when $w_{\pi,j} \ne 0$, and zero otherwise. Extend the selected weights over their chain intervals by

$$c_n(u) = \sum_{\pi,j} W_\pi \omega_{\pi,j} \mathbf{1}_{\{|w_{\pi,j}| > uW_\pi\}} \mathbf{1}_{\{n_{j-1} < n \le n_j\}}.$$

Within one chain the intervals are disjoint, and

$$\#\{j : |w_{\pi,j}| > uW_\pi\} \le u^{-q}.$$

Every interval has two endpoint changes. Hence

$$\max_n |c_n(u)| \le W, \qquad \sum_n |c_{n+1}(u) - c_n(u)| \le 2u^{-q}W. \tag{3.2}$$

The elementary identity $w_{\pi,j} = \int_0^1 W_\pi \omega_{\pi,j} \mathbf{1}_{\{|w_{\pi,j}|>uW_\pi\}}\,du$ and expansion into adjacent differences give

$$\sum_{\pi,j} \langle z_{\pi,j}, w_{\pi,j}\rangle = \int_0^1 \Lambda_\gamma(F,G;c(u))\,du.$$

Theorem 2.3 bounds its absolute value by

$$C\|F\|_3\|G\|_3\|W\|_3 \int_0^1 u^{-q/2}\,du = \frac{2C}{2-q}\|F\|_3\|G\|_3\|W\|_3.$$

Since $2-q=(r-2)/(r-1)$, this proves Theorem 2.1 for a finite list. The integral reconstruction is justified by $\sum_{\pi,j}|z_{\pi,j}w_{\pi,j}| \le V^r(b_n:n)W \in L^1$ on a finite list.

Monotone convergence over finite blocks handles a countable list.

4. The energy inequality from the heat equation

Fix $u(x,y), v(x,z) \in L^3(\mathbb{R}^2)$. At scale $s>0$ and centres $p=(p_0,p_1,p_2)$, define

$$C(x,y) = u(x,y)\sqrt{\rho_{s/6}(x-p_0)\rho_s(y-p_1)}, \qquad D(x,z) = v(x,z)\sqrt{\rho_{s/6}(x-p_0)\rho_s(z-p_2)}.$$

These are Hilbert–Schmidt kernels. Put $R=[C,D]$, $A=CC^*$, $B=DD^*$, $T=R^*R$, and $w_s(p)=\rho_{5s/6}(p_0+p_1+p_2)$. For $U\ge 0$ write $\Phi(U)=\operatorname{tr}(U^{3/2})$ and define one energy:

$$E_s = \int_{\mathbb{R}^3} \left[\Phi(A+B) + \tfrac12\Phi(A) + \tfrac12\Phi(B)\right] w_s(p)\,dp. \tag{4.1}$$

The unused centre integrates out in each of the last two terms. Jensen, $\Phi(CC^*) \le \|C\|_{\mathrm{HS}}^3$, and $\Phi(A+B) \le \sqrt{2}[\Phi(A)+\Phi(B)]$ imply

$$0 \le E_s \le (\tfrac12+\sqrt{2})(\|u\|_3^3 + \|v\|_3^3). \tag{4.2}$$

The quadratic form below is the Hessian of Φ (with the regularization in Appendix A at zero spectrum):

$$\mathfrak{b}_U(K) = \frac{3}{2}\int_0^\infty \|e^{-t\sqrt{U}/2}Ke^{-t\sqrt{U}/2}\|_{\mathrm{HS}}^2\,dt, \qquad K=K^*.$$

We shall prove

$$\boxed{\frac{1}{12}\int_a^b \int_{\mathbb{R}^3} \mathfrak{b}_{T_s}(\partial_{p_0}T_s) w_s(p)\,dp\,ds \le E_a - E_b.} \tag{4.3}$$

4.1. **The derivative calculation.** Lemma A.1 proves the Gaussian chain rule below for the given L^3 inputs. Scalar trace derivatives are weak derivatives; their centre integrals converge absolutely, and E_s is locally absolutely continuous. All the following identities therefore hold in their integrated form as well. The idea is to differentiate the main term first, then use the two separate energies to cancel its only positive error.

Put $A_1=\partial_{p_1}A$, $B_2=\partial_{p_2}B$, and $f=\Phi(A+B)=\Phi(T)$. For an outer variable use $A+B$, which is linear in its Gaussian density; for the common variable use T, which is linear in the common Gaussian density. The heat chain rule contributes common curvature with coefficient $1/12$ and each outer curvature with coefficient $1/2$.

After integration by parts, the Gaussian second derivatives and $\partial_s w_s$ together contribute $5/12 + 1/12 + 1/2 + 1/2 = 3/2$ times $\int f w_s''$. Here $w_s'' = \rho''_{5s/6}(p_0+p_1+p_2)$, and

$$\int f w_s''\,dp = \int \partial_{p_1}\partial_{p_2} f\; w_s\,dp.$$

Consequently,

$$\frac{d}{ds}\int f w_s\,dp + \frac{1}{12}\int \mathfrak{b}_T(\partial_{p_0}T)w_s\,dp = \int\left[\frac{3}{2}\partial_{p_1}\partial_{p_2}f - \frac{\mathfrak{b}_{A+B}(A_1)+\mathfrak{b}_{A+B}(B_2)}{2}\right]w_s\,dp.$$

Since A is independent of p_2 and B of p_1, the mixed chain rule and Cauchy–Schwarz for the nonnegative Hessian give

$$2\partial_{p_1}\partial_{p_2}f \le \mathfrak{b}_{A+B}(A_1)+\mathfrak{b}_{A+B}(B_2).$$

The order property (A.2) then yields

$$\frac{d}{ds}\int f w_s\,dp + \frac{1}{12}\int \mathfrak{b}_T(\partial_{p_0}T)w_s\,dp \le \frac{1}{4}\int[\mathfrak{b}_A(A_1)+\mathfrak{b}_B(B_2)]w_s\,dp. \tag{4.4}$$

The two separate energies cancel this error. Indeed, first integrate out the unused centre in $\int\Phi(A)w_s$ and apply the heat chain rule in the remaining centres, using $\Phi(A)=\Phi(C^*C)$ for the common variable. Its contribution is nonpositive, and the same argument applies to B. Thus

$$\frac{d}{ds}\int\Phi(A)w_s\,dp \le -\frac{1}{2}\int\mathfrak{b}_A(A_1)w_s\,dp, \qquad \frac{d}{ds}\int\Phi(B)w_s\,dp \le -\frac{1}{2}\int\mathfrak{b}_B(B_2)w_s\,dp.$$

Adding half of each inequality to (4.4) gives exactly

$$\boxed{E_s' + \frac{1}{12}\int\mathfrak{b}_T(\partial_{p_0}T)w_s\,dp \le 0.} \tag{4.5}$$

Integrating from a to b proves (4.3). This explains the coefficient $1/2$ in the energy: each separate term supplies dissipation with coefficient $1/2$, whereas the main term leaves an error with coefficient $1/4$. The factor $1/12$ comes from the common variance $s/6$ and the heat-equation factor $1/2$. The centre variance $5s/6$ is also exactly the variance in the convolution identity used below.

5. From the Gaussian pairing to the curvature integral

Put $s_n = t_n^2/(2\pi)$ and $\theta(t)=(1-t)_+^2$. For $0<a<b$,

$$\rho_b-\rho_a = \frac{1}{16}\int_0^\infty[\theta(a/s)-\theta(b/s)]s^2\rho_s^{(6)}\,ds. \tag{5.1}$$

Indeed, $w(s)=s^2[\theta(a/s)-\theta(b/s)]$ satisfies $w'''=2(\delta_a-\delta_b)$, while $\rho_s^{(6)}=8\partial_s^3\rho_s$. Three integrations by parts give the identity. The boundary terms vanish in L^1: for large s, $w=O(s)$, $w'=O(1)$, and $w''=0$, whereas $\|\partial_s^j\rho_s\|_1 \lesssim s^{-j}$ for $j=1,2$. The two θ terms are kept together; their difference makes the scale integral absolutely convergent in L^1.

Define one continuous coefficient path by

$$c_s = \sum_{n=1}^{N} c_n[\theta(s_{n-1}/s)-\theta(s_n/s)]. \tag{5.2}$$

Its weights are nonnegative and sum to $\theta(s_0/s)-\theta(s_N/s)\le 1$. Hence $|c_s|\le H$. Summation by parts gives

$$c_s' = \sum_{n=0}^{N}(c_{n+1}-c_n)\frac{2s_n}{s^2}(1-s_n/s)_+.$$

The path is continuously differentiable in L^3 and vanishes for $s\le s_0$. The scalar derivative weights are nonnegative, each has scale integral one, and each is at most $1/(2s)$. Thus

$$|c_s|\le H, \qquad \int_0^\infty|c_s'|\,ds \le V, \qquad |c_s'|\le\frac{V}{2s}. \tag{5.3}$$

Tonelli gives $\int_0^\infty [\theta(a/s) - \theta(b/s)]\, ds/s = \log(b/a)$; one may use $\theta(a/s) - \theta(b/s) = \int_a^b (2/s)(1 - u/s)_+\, du$. Consequently,

$$\int_{s_0}^\infty \|c_s\|_3 \, \frac{ds}{s} \le \|H\|_3 \log(s_N/s_0) < \infty.$$

Since $\gamma_t = \rho_{t^2/(2\pi)}$, define the sheared functions

$$h_s(a,b) = \overline{c_s(-a,-b)}, \qquad f(b,c) = F(b+c,-b), \qquad g(c,a) = G(-a,a+c).$$

Apply (5.1) to every Gaussian difference. The determinant-one shear gives

$$\Lambda_\gamma(F,G;c) = \frac{1}{16}\int_{s_0}^\infty \iiint h_s(a,b) f(b,c) g(c,a) s^3 \rho_s^{(6)}(a+b+c)\, da\, db\, dc\, \frac{ds}{s}. \tag{5.4}$$

The preceding scale bound, $\|\rho_s^{(6)}\|_1 \lesssim s^{-3}$, and Hölder at each translation justify every interchange. The logarithm is used only for absolute convergence, not in the uniform coefficient estimate.

Define

$$\psi = s\rho''_{s/3}, \qquad \psi * \psi * \psi = s^3 \rho_s^{(6)}.$$

For centres on $\Pi = \{p_0 + p_1 + p_2 = 0\}$, with measure $dp = dp_0 dp_1$, put $q_j(x) = \rho_s(x - p_j)$ and define three kernels:

$$W_0(a,b) = h_s(a,b)\sqrt{q_0(a) q_1(b)},$$
$$W_1(b,c) = f(b,c)\sqrt{q_1(b) q_2(c)},$$
$$W_2(c,a) = g(c,a)\sqrt{q_2(c) q_0(a)}.$$

With $\eta_v(x) = \psi(x - p_v)/q_v(x)$ we have $\max_v \|\eta_v\|_\infty \le 3\sqrt{3}$ and

$$\tau_s(p) = \operatorname{tr}([\eta_0] W_0 [\eta_1] W_1 [\eta_2] W_2), \qquad \Lambda_\gamma = \frac{1}{16}\int_{s_0}^\infty \int_\Pi \tau_s(p)\, dp\, \frac{ds}{s}.$$

Here $[\eta]$ denotes multiplication by η, indices are read modulo three, and an unsubscripted operator norm is the L^2 operator norm. The second identity is Gaussian convolution in (5.4). Fubini follows from

$$\iiint |h(a,b) f(b,c) g(c,a) k(a+b+c)|\, da\, db\, dc \le \|k\|_1 \|h\|_3 \|f\|_3 \|g\|_3,$$

proved by undoing the shear and applying Hölder at each translation. The same estimate applies to $|\psi| * |\psi| * |\psi|$; its L^1 norm is bounded uniformly in s.

For these three functions form the cyclic row pairs and their energies $E_{v,s}$ as in Section 4: for each v, use variance $s/6$ in the shared variable and variance s in the other two variables. Put $Q_{v,s} = \mathfrak{b}_{T_{v,s}}(\partial_{p_v} T_{v,s})$, where $T_{v,s}$ is the correlation with these variances. The coefficient c_s changes with scale. Section 6 will account for its derivative when integrating the heat inequality.

5.1. **An estimate for the trace of three operators.** For three kernels and bounded real multipliers, set

$$\tau = \operatorname{tr}([\eta_0] W_0 [\eta_1] W_1 [\eta_2] W_2), \qquad R_v = [W_v, W_{v-1}^*].$$

We claim that

$$|\tau| \le \frac{1}{\sqrt{3}} \max_v \|\eta_v\|_\infty \sum_v \mathfrak{b}_{R_v^* R_v}(R_v^* [\eta_v] R_v). \tag{5.5}$$

First, for any selfadjoint Hilbert–Schmidt operator M and bounded selfadjoint operator D, we have

$$|\operatorname{tr}((DM)^3)| \le 2\|D\|\, \mathfrak{b}_{M^2}(MDM). \tag{5.6}$$

To prove this, put $Q = |M|$ and $\mathcal{F}(t) = \operatorname{tr}((DMe^{-tQ})^3)$. Differentiation and trace cyclicity give

$$\begin{aligned}|\mathcal{F}'(t)| &= 3\big|\operatorname{tr}(QMe^{-tQ}DMe^{-tQ}DMe^{-tQ}D)\big| \\ &\le 3\|D\|\,\|Me^{-tQ}DQe^{-tQ/2}\|_{\mathrm{HS}}^2 \\ &\le 3\|D\|\,\|Qe^{-tQ/2}DQe^{-tQ/2}\|_{\mathrm{HS}}^2.\end{aligned}$$

For the first inequality, use $QMe^{-tQ} = (\operatorname{sgn} M)(Qe^{-tQ/2})^2$, move one factor $Qe^{-tQ/2}$ around the trace, and apply Hilbert–Schmidt Cauchy–Schwarz. The second follows from $(Me^{-tQ})^2 \le (Qe^{-tQ/2})^2$. Integrate on $[0,T]$. Since $\|Me^{-TQ}\| \le 1/(eT)$ and $\|Me^{-TQ}\|_{\mathrm{HS}} \le \|M\|_{\mathrm{HS}}$, we have $\mathcal{F}(T) \to 0$. The definition of $\mathfrak{b}$ and the polar decomposition of M give

$$\int_0^\infty \|Qe^{-tQ/2}DQe^{-tQ/2}\|_{\mathrm{HS}}^2\,dt = \frac{2}{3}\mathfrak{b}_{M^2}(MDM).$$

Letting $T \to \infty$ proves (5.6). This argument allows an infinite value on the right; no separate integrability estimate is needed.

Now multiply one W_v by a constant of modulus one so that $\tau = |\tau|$. This only conjugates the affected correlations and directions by unitaries, and hence preserves the right side of (5.5). Let $M = M^*$ have blocks $M_{v,v+1} = W_v$, the adjoint blocks in the reverse direction, and zero diagonal blocks, and put $D = \operatorname{diag}([\eta_v])$. Then $\operatorname{tr}((DM)^3) = 6|\tau|$. If 1_v projects onto the vth copy, joint convexity and the homogeneity $\mathfrak{b}_{\lambda A}(\lambda K) = \lambda^{3/2}\mathfrak{b}_A(K)$ give

$$\begin{aligned}6|\tau| &\le 2\|D\|\,\mathfrak{b}_{M^2}(MDM) \\ &\le 2\sqrt{3}\|D\|\sum_v \mathfrak{b}_{M1_vM}(M1_vDM) \\ &= 2\sqrt{3}\max_v\|\eta_v\|_\infty \sum_v \mathfrak{b}_{R_v^*R_v}(R_v^*[\eta_v]R_v).\end{aligned}$$

The last equality identifies the two nonzero blocks in the vth row of M with R_v. Dividing by six proves (5.5).

5.2. **Gaussian convolution and the curvature estimate.** At the original configuration all three variances are s. For a fixed index v write $R = [W_v, W_{v-1}^*]$, $T = R^*R$, and $K = R^*[\eta_v]R$. Replace its common variance s by $s/6$ and translate the common centre by d; write T_d for this correlation. Gaussian convolution gives

$$T = \int T_d\rho_{5s/6}(d)\,dd, \qquad K = -s\int \partial_{p_v}T_d\,\rho'_{s/6}(d)\,dd.$$

The second identity follows from $\rho'_{s/6} * \rho'_{s/6} = \rho''_{s/3}$; the minus sign comes from differentiation in the centre. Direct Gaussian comparison gives

$$|s\rho'_{s/6}(d)| \le 4\sqrt{s}\,\rho_{5s/6}(d).$$

Joint convexity of $\mathfrak{b}$ and its quadratic dependence on the direction, followed by the change from $p_0+p_1+p_2 = 0$ and d to three unrestricted centres, yields

$$\int_{\{p_0+p_1+p_2=0\}} \mathfrak{b}_T(K)\,dp \le 16s\int_{\mathbb{R}^3} Q_{v,s}(p)\rho_{5s/6}(p_0+p_1+p_2)\,dp. \tag{5.7}$$

The measure on the plane is dp_0dp_1, with $p_2 = -p_0-p_1$; the stated change of variables has absolute Jacobian one. The density on the right is exactly the one in the energy inequality.

5.3. **Combining the estimates.** Equations (5.5) and (5.7) give

$$|\Lambda_\gamma(F,G;c)| \le 3\sum_v \int_{s_0}^{\infty}\int_{\mathbb{R}^3} Q_{v,s}(p)\rho_{5s/6}(p_0+p_1+p_2)\,dp\,ds. \tag{5.8}$$

The next section integrates the heat inequality with the changing coefficient c_s and completes the coefficient estimate.

6. Completing the coefficient estimate

Proof of Theorem 2.3. Let $E_s = \sum_v E_{v,s}$ be the sum of the three energies formed from h_s, f, g. In its derivative, the Gaussian factors give the negative curvature term from Section 4. We estimate the remaining terms, which come from differentiating h_s.

Let M_1, M_2 be the centred one-dimensional maximal operators in the original coordinates, and set

$$H_i = (M_i|H|^2)^{1/2}, \qquad F_1 = (M_1|F|^2)^{1/2}, \qquad G_2 = (M_2|G|^2)^{1/2}.$$

Their L^3 norms are bounded by an absolute constant times the L^3 norms of the corresponding functions. We claim that

$$\frac{d}{ds}E_s + \frac{1}{12}\sum_v \int Q_{v,s}(p)\rho_{5s/6}(p_0+p_1+p_2)\,dp \le C\int |c_s'|\,[H_1H_2 + H_1G_2 + H_2F_1]. \tag{6.1}$$

Only two pairs contain the coefficient. For either one, write $R = [C, D]$ with

$$C(x,y) = h_s(x,y)\sqrt{q_0(x)q_1(y)}, \qquad D(x,z) = v(x,z)\sqrt{q_0(x)q_2(z)},$$

using the coefficient or its adjoint as needed. Here $q_0 = \rho_{s/6}(\cdot - p_0)$ and $q_i = \rho_s(\cdot - p_i)$ for $i = 1, 2$. The part of the derivative of $\Phi(R^*R)$ due to h_s is

$$3\Re \iint \overline{(R\sqrt{R^*R})(x,y)}\,\dot h_s(x,y)\sqrt{q_0(x)q_1(y)}\,dx\,dy.$$

The row and column Cauchy–Schwarz inequality gives

$$|(R\sqrt{R^*R})(x,y)| \le \|R(x,\cdot)\|_2\|R(\cdot,y)\|_2.$$

Write $\widetilde H$ for H in these coordinates. At fixed x, y, put

$$\begin{aligned} A(p_1) &= \int |\widetilde H(x,y')|^2 q_1(y')\,dy',\\ B(p_0) &= \int |\widetilde H(x',y)|^2 q_0(x')\,dx',\\ D(p_2) &= \int |v(x,z')|^2 q_2(z')\,dz'. \end{aligned}$$

Since $|h_s| \le \widetilde H$, the absolute derivative integrand is at most $3|\dot h_s| q_0q_1\sqrt{B}\sqrt{A+D}$. The centre measure

$$d\mu = q_0(x)q_1(y)\rho_{5s/6}(p_0+p_1+p_2)\,dp$$

has mass one, and direct Gaussian convolution gives

$$\begin{aligned} \int A\,d\mu &= (\rho_{2s} *_2 |\widetilde H|^2)(x,y),\\ \int B\,d\mu &= (\rho_{s/3} *_1 |\widetilde H|^2)(x,y),\\ \int D\,d\mu &= (\rho_{3s} *_2 |v|^2)(x,-x-y). \end{aligned} \tag{6.2}$$

For the last identity the centre variance is $s/6 + s + 5s/6 = 2s$, and the kernel adds s. Cauchy–Schwarz in μ bounds the integral of $\sqrt{B}\sqrt{A+D}$ by

$$\left(\int B\,d\mu\right)^{1/2}\left[\left(\int A\,d\mu\right)^{1/2}+\left(\int D\,d\mu\right)^{1/2}\right].$$

Each Gaussian average is bounded by the corresponding maximal function. Undoing the shear gives $H_1H_2 + H_1G_2$ for the pair with fixed function g, and $H_1H_2 + H_2F_1$ for the pair with fixed function f. The separate coefficient energies contribute only H_1H_2. This proves (6.1).

The Gaussian and coefficient chain rules are justified in Lemma A.1. Indeed c_s is continuously differentiable in L^3, and (5.3) gives $|c_s'| \le V/(2s)$. Thus all contributions are integrable on compact positive scale intervals. The coefficient contribution is also integrable over the full scale range, by $\int |c_s'|\,ds \le V$ and Hölder.

Integrate (6.1) over $[s_0, R]$. Since $c_{s_0} = 0$, $E_{s_0} \lesssim \|F\|_3^3 + \|G\|_3^3$. Discard the nonnegative terminal energy and let $R \to \infty$. Using $\int |c_s'|\,ds \le V$ before applying Hölder, and using (5.8), gives

$$\begin{aligned}|\Lambda_\gamma(F,G;c)| \lesssim{}& \|F\|_3^3 + \|G\|_3^3\\ &+ \|V\|_3\|H\|_3(\|H\|_3 + \|F\|_3 + \|G\|_3).\end{aligned}\tag{6.3}$$

By trilinear homogeneity we may assume

$$\|F\|_3 = \|G\|_3 = 1, \qquad \|H\|_3\|V\|_3 = 1.$$

Since $V \ge 2H$, we have $\|H\|_3 \le 1/\sqrt{2}$, and the right side of (6.3) is bounded by an absolute constant. Rescaling proves $|\Lambda_\gamma| \lesssim \|F\|_3\|G\|_3\sqrt{\|H\|_3\|V\|_3}$. If a normalizing quantity is zero, the conclusion is immediate. □

Appendix A. The operator facts actually used

All kernels act on $L^2(\mathbb{R})$ or finitely many labelled copies. A square-integrable kernel is Hilbert–Schmidt; R^*R, RR^* are positive trace class and have the same nonzero spectrum. Hence $\Phi(R^*R) = \Phi(RR^*)$. Trace cyclicity is used only for trace-class products. No replacement by finite-rank kernels is made.

Curvature, order, and averaging. We write $\mathfrak{b}_A(K,L)$ for the polarization of $\mathfrak{b}_A$. On selfadjoint Hilbert–Schmidt operators, the positive operation $Z \mapsto \sqrt{A}Z + Z\sqrt{A}$ and completion of a quadratic square give

$$\mathfrak{b}_A(K) = \frac{3}{2}\sup_{Z=Z^*\in\mathcal{S}^2}\{2\,\mathrm{tr}(KZ) - 2\,\mathrm{tr}(\sqrt{A}\,Z^2)\}.\tag{A.1}$$

The formula is interpreted as an extended quadratic form; the supremum need not be attained. It follows as well by integrating its exponential semigroup in the definition of $\mathfrak{b}$. The exact identity

$$\sqrt{A} = \frac{1}{\pi}\int_0^\infty \lambda^{-1/2}A(A+\lambda I)^{-1}\,d\lambda$$

and inverse order of positive operators give

$$A \le B \quad\Longrightarrow\quad \sqrt{A} \le \sqrt{B}, \qquad \mathfrak{b}_B(K) \le \mathfrak{b}_A(K).\tag{A.2}$$

For a positive measure ν of total mass one, Cauchy–Schwarz gives $(\int \sqrt{A_d}\,d\nu)^2 \le \int A_d\,d\nu$ and therefore $\int \sqrt{A_d}\,d\nu \le \sqrt{\int A_d\,d\nu}$. Insert this into (A.1), first at a fixed Z, to obtain

$$\mathfrak{b}_{\int A_d\,d\nu}\left(\int K_d\,d\nu\right) \le \int \mathfrak{b}_{A_d}(K_d)\,d\nu.$$

Here it suffices that $\int \mathrm{tr}\,A_d\,d\nu$ and $\int \|K_d\|_{\mathrm{HS}}\,d\nu$ are finite. The Gaussian averages in the proof meet these conditions by Hölder and Gaussian moments. The same formula gives $\mathfrak{b}_{\lambda A}(\lambda K) = \lambda^{3/2}\mathfrak{b}_A(K)$.

Lemma A.1 (Differentiating the Gaussian trace integrals)**.** *For the Gaussian Gram operators in this proof, the trace functions have locally integrable weak first derivatives in variance and weak second derivatives in the centres. If U depends linearly on a Gaussian of variance σ and centre p, then*

$$(\partial_\sigma - \tfrac{1}{2}\partial_p^2)\Phi(U) = -\tfrac{1}{2}\mathfrak{b}_U(\partial_p U). \tag{A.3}$$

For $U = A(p_1) + B(p_2)$ the mixed derivative is

$$\partial_{p_1}\partial_{p_2}\Phi(A+B) = \mathfrak{b}_{A+B}(\partial_{p_1}A, \partial_{p_2}B). \tag{A.4}$$

For a continuously differentiable Hilbert–Schmidt family,

$$\frac{d}{d\theta}\Phi(R_\theta^* R_\theta) = 3\Re\langle R_\theta\sqrt{R_\theta^* R_\theta}, R_\theta'\rangle_{\mathrm{HS}}. \tag{A.5}$$

The centre-integrated energies are locally absolutely continuous in scale, and these formulas permit integration by parts and the scale and coefficient chain rule for the path c_s used here.

Proof. Put $\Phi_\eta(U) = \operatorname{tr} f_\eta(U)$, where

$$f_\eta(t) = (t+\eta)^{3/2} - \eta^{3/2} - \tfrac{3}{2}\sqrt{\eta}\,t, \qquad \eta > 0.$$

This is the trace of one trace-class expression; $f_\eta(0) = 0$ and $0 \le f_\eta(t) \uparrow t^{3/2}$ as $\eta \downarrow 0$. The square-root resolvent gives

$$D\Phi_\eta(U)[K] = \tfrac{3}{2}\operatorname{tr}((\sqrt{U+\eta I} - \sqrt{\eta}I)K), \qquad D^2\Phi_\eta(U)[K,L] = \mathfrak{b}_{U+\eta I}(K,L).$$

Indeed the quadratic second derivative equals $\frac{3}{2\pi}\int_0^\infty \lambda^{1/2}\operatorname{tr}(KR_\lambda K R_\lambda)\,d\lambda$, with $R_\lambda = (U + \eta I + \lambda I)^{-1}$. The ordinary chain rule proves the first two identities with Φ_η and $\mathfrak{b}_{U+\eta I}$.

For uniform domination write $U = \int K_z \rho_\sigma(z-p)\,dz$ with $K_z \ge 0$, and put $v = (z-p)/\sqrt{\sigma}$. These trace-class integrals and their Gaussian derivatives exist by $L^3 \cdot L^6 \subset L^2$. Jensen and joint convexity of $\mathfrak{b}$ give

$$\Phi_\eta(U) \le \int \Phi(K_z)\rho_\sigma(z-p)\,dz, \qquad \mathfrak{b}_{U+\eta I}(\partial_p U) \le \frac{3}{4\sigma}\int v^2\Phi(K_z)\rho_\sigma(z-p)\,dz.$$

Here $\mathfrak{b}_{K+\eta I}(K) \le 3\Phi(K)/4$. For positive K, Schatten Hölder and Young also give $|D\Phi_\eta(U)[K]| \le \frac{1}{2}\Phi(U) + \Phi(K)$. Insert the Gaussian derivatives and absorb polynomial factors by doubling the Gaussian variance. Uniformly in $\eta > 0$,

$$\begin{aligned} |\partial_p\Phi_\eta(U)| &\lesssim \sigma^{-1/2}\int \Phi(K_z)\rho_{2\sigma}(z-p)\,dz, \\ |\partial_\sigma\Phi_\eta(U)| + |\partial_p^2\Phi_\eta(U)| &\lesssim \sigma^{-1}\int \Phi(K_z)\rho_{2\sigma}(z-p)\,dz. \end{aligned} \tag{A.6}$$

The curvature satisfies the second bound as well.

For the common variable K_x is rank one and

$$\Phi(K_x) \le \sqrt{2}\left[\int |u(x,y)|^3\rho_s(y-p_1)\,dy + \int |v(x,z)|^3\rho_s(z-p_2)\,dz\right].$$

For an outer variable use $K_y + B$, where K_y is rank one, $\Phi(K_y) \le \int |u(x,y)|^3\rho_{s/6}(x-p_0)\,dx$, and $\Phi(K_y+B) \le \sqrt{2}[\Phi(K_y)+\Phi(B)]$. Each resulting scalar term omits one centre. Integrate that centre out of the centre-sum Gaussian, then integrate the remaining Gaussians. The integrals of first centre derivatives are at most $Cs^{-1/2}(\|u\|_3^3 + \|v\|_3^3)$; second derivatives and curvatures have the analogous Cs^{-1} bound. The same estimates hold with any positive centre-sum variance. They therefore also cover derivatives of its density and are uniform on compact positive scale intervals after one Gaussian enlargement. Centre cutoffs can consequently be removed in all the integration-by-parts identities.

The first trace derivatives converge to $\frac{3}{2}\operatorname{tr}(\sqrt{U}K)$ because $\|\sqrt{U+\eta I}-\sqrt{\eta}I-\sqrt{U}\|\le\sqrt{\eta}$. The quadratic forms increase to $\mathfrak{b}_U$. The bounds above give convergence in local L^1 of the derivative formulas, proving (A.3) for weak derivatives. For the mixed derivative use polarization for convergence and $|\mathfrak{b}_{U+\eta I}(A_1,B_2)|\le[\mathfrak{b}_{U+\eta I}(A_1)+\mathfrak{b}_{U+\eta I}(B_2)]/2$ for domination. This proves (A.4). For the scale identity, restrict the smooth regularized formulas to variances $(s/6,s,s)$ before passing to the limit. The preceding bounds give locally integrable weak scale derivatives of the energies. On compact positive scale intervals, fixed Gaussians of larger variance dominate the energy integrands, so dominated convergence also shows that the integral-defined energies are continuous. They are therefore the locally absolutely continuous representatives just obtained.

For the kernel derivative put $B_\eta(P)=\sqrt{P+\eta I}-\sqrt{\eta}I$. The regularized derivative is

$$\frac{d}{d\theta}\Phi_\eta(R_\theta^*R_\theta)=3\Re\langle R_\theta B_\eta(R_\theta^*R_\theta),R_\theta'\rangle_{\mathrm{HS}}.$$

Since $B_\eta(P)^2\le P$, its squared column norms are at most those of R; integrate over arbitrary measurable sets of columns to see this. Cauchy–Schwarz gives

$$|(RB_\eta(R^*R))(x,y)|\le\|R(x,\cdot)\|_2\|R(\cdot,y)\|_2. \tag{A.7}$$

Moreover $\|R(B_\eta(R^*R)-\sqrt{R^*R})\|_{\mathrm{HS}}\le\sqrt{\eta}\|R\|_{\mathrm{HS}}$. Integrate the derivative, pass to the limit, and use continuity of the limiting integrand to get (A.5). The pointwise bound holds in the limit as well.

Finally c_s is continuously differentiable in L^3, with $|c_s|\le H$ and $|c_s'|\le V/(2s)$ by (5.3). Its value and derivative therefore have fixed L^3 bounds on compact positive scale intervals. The pointwise estimate above, integrated as in Section 6, dominates the coefficient derivative. Apply the regularized scale and coefficient chain rule and pass to the limit using this bound and (A.6). This proves its integrated form for the original inputs, without approximation. □

Appendix B. Sharp averages at lacunary times

We compare the sharp kernel with the Gaussian at the prescribed times. Throughout this appendix, $b=\mathbf{1}_{[0,1]}$, $g(u)=e^{-\pi u^2}$, $\chi=g'/\sqrt{\pi}$, and $\chi^a=\chi(\cdot-a)$. The only square estimate needed is

$$\left(\sum_n\|T^{\chi^a}_{\tau t_n}(F,G)\|_2^2\right)^{1/2}\le C_\lambda(1+|a|)^{1/2}\|F\|_4\|G\|_4,\qquad\tau>0. \tag{B.1}$$

We first prove this estimate for real Schwartz inputs; the estimates then extend to complex L^4 inputs by splitting into real and imaginary parts and by density.

A continuous square estimate. Put $P=\|F\|_4\|G\|_4$. Uniformly for $A>0$, we claim

$$\int_{\mathbb{R}}g_A(a)\int_0^\infty\|T_t^{\chi^a}(F,G)\|_2^2\,\frac{dt}{t}\,da\le 5P^2. \tag{B.2}$$

Expand the square. With $\Sigma=\xi+\eta$, $\Delta=\xi-\eta$, and $B=1+2A^2$, its kernel multiplier is

$$\int_0^\infty\widehat{\chi}(t\xi)\widehat{\chi}(t\eta)e^{-\pi A^2t^2\Sigma^2}\,\frac{dt}{t}=-\frac{2\xi\eta}{\xi^2+\eta^2+A^2\Sigma^2}=1-\frac{(B+1)\Sigma^2}{B\Sigma^2+\Delta^2}.$$

The constant term contributes $\int F^2G^2\le P^2$. Let Φ_B be the positive Gaussian density with

$$\widehat{\Phi}_B(\xi,\eta)=e^{-\pi(B\Sigma^2/4+\Delta^2/2)},\qquad\varrho=\frac{\sqrt{B}\,t}{2\sqrt{2}}.$$

Direct integration gives

$$\frac{(B+1)\Sigma^2}{B\Sigma^2+\Delta^2}=-\frac{2(B+1)}{B}\int_0^\infty\widehat{\Phi}_B(t\xi,t\eta)\widehat{\chi}(\varrho\Sigma)^2\,\frac{dt}{t}.$$

Set $f(y,u) = F(u-y,y)$ and $k(x,u) = G(x,u-x)$, and write $\Phi_{B,t}(u,v) = t^{-2}\Phi_B(u/t, v/t)$. The four-linear form inside the last scale integral is

$$\int_{\mathbb{R}^4} \Phi_{B,t}(u-p-q, v-p-q) \left[\int_{\mathbb{R}} f(y,u)f(y,v)\chi_\varrho(y-q)\,dy\right] \times \left[\int_{\mathbb{R}} k(x,u)k(x,v)\chi_\varrho(x-p)\,dx\right] du\,dv\,dp\,dq.$$

Since $\Phi_{B,t} \geq 0$ and $\int \Phi_{B,t}(u-p, v-p)\,dp = g_{\sqrt{2}t}(u-v)$, Cauchy–Schwarz, including the scale integral, bounds its absolute value by $\Gamma(f)^{1/2}\Gamma(k)^{1/2}$, where

$$\Gamma(f) = \int_0^\infty \int_{\mathbb{R}^3} g_{\sqrt{2}t}(u-v) \left[\int_{\mathbb{R}} f(x,u)f(x,v)\chi_\varrho(x-q)\,dx\right]^2 du\,dv\,dq\,\frac{dt}{t}.$$

To bound this square integral, define

$$E_f(t) = \int_{\mathbb{R}^4} f(x,u)f(y,u)f(x,v)f(y,v)g_{\sqrt{2}t}(u-v)g_{\sqrt{B}t/2}(x-y)\,dx\,dy\,du\,dv.$$

Indeed,

$$\int_{\mathbb{R}} \chi_\varrho(x-q)\chi_\varrho(y-q)\,dq = -t\partial_t g_{\sqrt{2}\varrho}(x-y).$$

Differentiating the two Gaussian factors therefore gives

$$-tE_f'(t) = D_x(t) + D_u(t), \qquad \int_0^\infty D_x(t)\,\frac{dt}{t} = \Gamma(f).$$

Both terms are nonnegative: each is the integral of a square against the other Gaussian, as in the displayed formula for Γ. Also $E_f(t) \geq 0$, by positive definiteness of the Gaussian, and $E_f(0+) = \|f\|_4^4$. Consequently $\Gamma(f) \leq \|f\|_4^4$. The same holds for k, and $2(B+1)/B \leq 4$ proves (B.2).

Sampling the continuous estimate. For $1/2 \leq \beta \leq 3/4$, $\chi = \beta^{-1}\chi_\beta * g_{\sqrt{1-\beta^2}}$. Jensen therefore gives

$$\|T_t^{\chi^a}(F,G)\|_2^2 \leq \beta^{-2} \int_{\mathbb{R}} \beta g_{\sqrt{1-\beta^2}}(\beta z - a)\|T_{\beta t}^{\chi^z}(F,G)\|_2^2\,dz.$$

Uniformly in this range of β, elementary Gaussian comparison yields

$$\beta g_{\sqrt{1-\beta^2}}(\beta z - a) \leq C(1+|a|)g_{8(1+|a|)}(z).$$

Average in $d\beta/\beta$ and sum over $t = \tau t_n$. For a 2-lacunary sequence the intervals $[\tau t_n/2, 3\tau t_n/4]$ are disjoint, so (B.2) proves (B.1). A λ-lacunary sequence splits into $\max\{1, \lceil \log 2/\log\lambda \rceil\}$ such sequences.

One integral for the sharp error. The following identity holds in $L^1(\mathbb{R})$:

$$b - g = -\sqrt{\pi}\int_0^1 (1-v)\chi^v\,dv - \frac{1}{2\sqrt{\pi}}\int_0^1 \left[\chi_\tau - (\chi^{1/\tau})_\tau\right] d\tau. \tag{B.3}$$

The first integral is $b*g - g$. For the second, use $\partial_\tau g_\tau = \tau g_\tau''/(2\pi)$ and $b' = \delta_0 - \delta_1$ to obtain

$$\partial_\tau(b * g_\tau) = \frac{1}{2\sqrt{\pi}}[\chi_\tau - \chi_\tau(\cdot - 1)].$$

This derivative is uniformly bounded in L^1, and $b*g_\tau \to b$ in L^1 as $\tau \downarrow 0$, proving the identity. Apply Minkowski and (B.1) to (B.3). The only improper integral is harmless: $\int_0^1 [1 + \sqrt{1+\tau^{-1}}]\,d\tau < \infty$. Thus, with $E_n = \mathcal{A}_{t_n} - \mathcal{G}_{t_n}$,

$$\|(E_n(F,G))_n\|_{L^2(\ell^2)} \leq C_\lambda \|F\|_4\|G\|_4. \tag{B.4}$$

Interpolation and variation. Directional maximal functions give, for every $p > 2$,

$$\sup_n |E_n(F,G)| \le C(M_x|F|^2)^{1/2}(M_y|G|^2)^{1/2}, \qquad \|(E_n(F,G))_n\|_{L^{p/2}(\ell^\infty)} \le C_p\|F\|_p\|G\|_p.$$

For $r > 3$, take $\theta = 2/r$ and $p = 6(r-2)/(2r-3) > 2$. Then $1/3 = (1-\theta)/p + \theta/4$ and $1/r = \theta/2$. Bilinear complex interpolation with (B.4) yields

$$\|(E_n(F,G))_n\|_{L^{3/2}(\ell^r)} \le C_{\lambda,r}\|F\|_3\|G\|_3.$$

Since $V^r(e_n : n) \le 2\|(e_n)_n\|_{\ell^r}$, Theorem 2.1 and the triangle inequality give the sharp-average variation bound. All constants are uniform over finite lists; monotone convergence handles an infinite sequence. The requirement $r > 3$ is precisely $p > 2$ in this interpolation, so this comparison does not cover $2 < r \le 3$.

References

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